Weixiang Yan

Senior Research Scientist, Apple Foundation Models | Cupertino, CA, USA
yanweixiang.ywx@gmail.com | +1 (603) 501-9963 | Homepage | Google Scholar

My research focuses on building LLM agents that can write code, operate computers, and serve as proactive personal assistants.

Education

University of California, Santa Barbara — *Santa Barbara, CA*
Master of Science in Computer Science — *Sept 2023 – Dec 2024*

Beijing University of Posts and Telecommunications — *Beijing, China*
Bachelor of Science in Computer Science and Bachelor of Science in Management — *Sept 2018 – Jun 2022*

Experience

Apple, Foundation Model Team — *Cupertino, CA*
Senior Research Scientist — *Dec 2025 – Present*

- **Computer Use RL (iOS & macOS):** Lead end-to-end agentic RL for UI navigation on iOS and macOS, reaching performance competitive with frontier LLMs.
  - **Training recipe & infrastructure:** Designed a multi-turn RL recipe with hybrid rewards, and built training and rollout infrastructure that runs 20K+ concurrent iOS and macOS environments.
  - **Environments & persona scaling:** Built 500+ first- and third-party app/web environments and scaled to 1K+ synthetic user personas to diversify tasks and user contexts.
  - **Data synthesis & evaluation:** Built task and trajectory synthesis pipelines for agent training, and an evaluation suite spanning internal and open-source benchmarks.
- **Agentic Coding Post-training:** Own end-to-end post-training of agentic coding for Apple Foundation Models, taking the capability from zero to strong benchmark performance.
  - **Data synthesis:** Built large-scale SWE-bench-style data synthesis pipelines for mid-training, SFT, and RL.
  - **Training recipe:** Designed post-training recipes combining SFT, on-policy distillation, and RL.
  - **Evaluation:** Built the coding evaluation framework, with safeguards against model reward hacking and sandbox escape.
- **Agent Sandbox Infrastructure:** Built from scratch a secure, multi-tenant, Docker-based coding sandbox on Kubernetes that supports 100K+ concurrent containers for RL rollouts, data synthesis, and large-scale evaluation.

Amazon, Generative Foundation Modeling Team — *Palo Alto, CA*
Applied Scientist II — *Dec 2024 – Dec 2025*

- **Hallucination Mitigation:** Responsible for end-to-end hallucination mitigation for the team, covering post-training data cleaning, evaluation framework development, RLVR-based training, and retrieval quality control; reduced the internal model's hallucination rate from 63.19% to 17.82% (a 72% relative reduction).
- **Agentic Coding RL:** Developed function-level and repository-level code RL training recipes, tackling low training efficiency, sparse rewards, and reward hacking; improved LiveCodeBench v6 from 59.6% to 76.4% and SWE-bench Verified from 50.6% to 65.6%.
- **Agentic Application RL:** Developed a hierarchical rubric-driven RLAIF reward and combined it with RLVR; the resulting application RL training recipe was deployed and improved the online model's helpfulness.
- **Helpfulness Evaluation:** Designed and built a helpfulness benchmark for multi-turn dialogues across diverse shopping scenarios, driven jointly by rubrics and LLM-as-a-Judge.

Alibaba Qwen Team — *Hangzhou, China*
Research Intern (Mentors: Wen Wang, Qian Chen) — *Jul 2022 – Jan 2023*

- **CodeScope (ACL 2024):** Proposed an execution-based, multilingual, multitask benchmark for LLM code understanding and generation across 43 programming languages and 8 coding tasks, and built MultiCodeEngine, an automated execution engine for 14 languages (100+ citations).
- **CodeTransOcean (EMNLP 2023):** Developed multilingual modeling approaches for code translation and demonstrated their great potential in improving the translation quality of both low-resource and high-resource language pairs (130+ citations).

- **Outline-Conditioned Generation (EACL 2024):** Leveraged the duality between summarization and generation to improve outline-conditioned generation, and proposed a unified outline-control approach to mitigate imbalanced outline utilization.
- **DialogueAttention (Chinese Patent CN116186244B, granted 2026):** Proposed an attention mechanism that highlights essential spans in long multi-party dialogues to guide summarization and reduce context noise.

Microsoft Research Asia, System Team — *Beijing, China*
Research Intern (Mentor: Yuanchun Li) — *Jun 2021 – Oct 2021*

- **WhyGen (ICSE 2022):** Built a tool that explains generated code by referring to the training examples behind it, notifying users of possible recitation and imitation with 81.21% top-10 accuracy.

Peking University, Department of Computer Science — *Beijing, China*
Research Intern (Mentor: Yao Guo) — *Nov 2020 – May 2021*

- **DependEX (ISSRE 2021):** Combined CNN-based semantic features with transformer-based context modeling to identify dependencies between UI form elements (over 92% accuracy on large-scale mobile web datasets), enabling automatic form filling and natural-language test case generation.

Papers ** equal contribution*

[20] **LLM Value Functions for Credit Assignment in Agentic RL** [*arXiv*]
Kaijie Zhu, **Weixiang Yan**, Omar Attia, Abbas Kazerouni, Alexander Toshev, Andrew Szot.

[19] **Demystify On-Policy Distillation Instability** [*Under review*]
Fengqing Jiang, Rui Meng, Chao Jiang, **Weixiang Yan**, . . . , Radha Poovendran, Meng Cao, Simon Wang.

[18] **HSearch: Harnessed Search Agents for Visual Web Research** [*Under review*]
Le Zhang, **Weixiang Yan**, Kaixin Ma, Haiming Gang, Afshin Dehghan, Mingfei Gao.

[17] **KEPO: Knowledge-Enhanced Preference Optimization for Multimodal Reasoning with Applications to Medical VQA** [*Under review*]
Fan Yang, Rui Meng, **Weixiang Yan**, Trudi Di Qi, Ali Ezzati, Yuxin Wen.

[16] **SearchCoder: Training Coding Agents for Information Seeking** [*Under review*]
Yukun Huang, **Weixiang Yan**, Dan Friedman, Haitian Sun, . . . , Bhuwan Dhingra.

[15] **AlphaQuanter: An E2E Tool-Augmented Agentic Reinforcement Learning Framework for Stock Trading** [*ACL 2026*]
Zheye Deng, **Weixiang Yan**, Changlong Yu, Jiashu Wang.

[14] **ClawsBench: Evaluating Capability and Safety of LLM Productivity Agents in Simulated Workspaces** [*COLM 2026*]
Xiangyi Li, Kyoung Whan Choe, . . . , **Weixiang Yan**, Yiyuan Li, Han-chung Lee.

[13] **CM2: Reinforcement Learning with Checklist Rewards for Multi-Turn and Multi-Step Agentic Tool Use** [*NeurIPS 2026*]
Zhen Zhang, Kaiqiang Song, Xun Wang, Yebowen Hu, **Weixiang Yan**, . . . , Xin Eric Wang, Song Wang.

[12] **Soft Thinking: Unlocking the Reasoning Potential of LLMs in Continuous Concept Space** [*NeurIPS 2025*]
Zhen Zhang, Xuehai He, **Weixiang Yan**, Ao Shen, Chenyang Zhao, . . . , Yelong Shen, Xin Eric Wang.

[11] **Think-RM: Enabling Long-Horizon Reasoning in Generative Reward Models** [*NeurIPS 2025*]
Ilgee Hong, Changlong Yu, Liang Qiu, **Weixiang Yan**, Zhenghao Xu, . . . , Chao Zhang, Tuo Zhao.

[10] **ClinicalLab: Aligning Agents for Multi-Departmental Clinical Diagnostics in the Real World** [*NeurIPS 2025*]
Weixiang Yan, Haitian Liu, Tengxiao Wu, Qian Chen, Wen Wang, . . .

[9] **CodeHalu: Investigating Code Hallucinations in LLMs via Execution-based Verification** [*AAAI 2025*]
Yuchen Tian*, **Weixiang Yan***, Qian Yang, Xuandong Zhao, Qian Chen, Wen Wang, . . . , Lei Ma, Dawn Song.

[8] **CodeJudge-Eval: Can Large Language Models be Good Judges in Code Understanding?** [*COLING 2025*]
Yuwei Zhao*, Ziyang Luo*, Yuchen Tian, Hongzhan Lin, **Weixiang Yan**, Annan Li, Jing Ma.

[7] **SpineBench: Benchmarking Multimodal LLMs for Spinal Pathology Analysis** [*ACM MM 2025*]
Chenghanyu Zhang, Zekun Li, Peipei Li, Xing Cui, Shuhan Xia, **Weixiang Yan**, Yiqiao Zhang, Qianyu Zhuang.

[6] **Decompose and Compare Consistency: Measuring VLMs' Answer Reliability via Task-Decomposition Consistency Comparison** [*EMNLP 2024*]
Qian Yang, **Weixiang Yan**, Aishwarya Agrawal.

[5] **CodeScope: An Execution-based Multilingual Multitask Multidimensional Benchmark for Evaluating LLMs on Code**

Understanding and Generation [*ACL 2024*]
Weixiang Yan, Haitian Liu, Yunkun Wang, Yunzhe Li, Qian Chen, Wen Wang, . . . , Hari Sundaram, Shuiguang Deng.

[4] **Advancing Precise Outline-Conditioned Text Generation with Task Duality and Explicit Outline Control** [*EACL 2024*]
Yunzhe Li, Qian Chen, **Weixiang Yan**, Wen Wang, Qinglin Zhang, Hari Sundaram.

[3] **CodeTransOcean: A Comprehensive Multilingual Benchmark for Code Translation** [*EMNLP 2023*]
Weixiang Yan, Yuchen Tian, Yunzhe Li, Qian Chen, Wen Wang.

[2] **WhyGen: Explaining ML-powered Code Generation by Referring to Training Examples** [*ICSE 2022*]
Weixiang Yan, Yuanchun Li.

[1] **Dependency-aware Form Understanding** [*ISSRE 2021*]
Shaokun Zhang, Yuanchun Li, **Weixiang Yan**, Yao Guo, Xiangqun Chen.

Service

Area Chair: ACL, ACL Rolling Review (ARR), AAAI
Reviewer: NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, AAAI, ARR, EACL, COLM

Honors & Awards

- Microsoft Research Asia “Stars of Tomorrow” Outstanding Intern Award
- National Scholarship (2020, 2021), WanDatong Scholarship, Outstanding Graduate, Outstanding Bachelor’s Thesis